Trajectory Optimization for Target Localization With Bearing-Only Measurement

Shaoming He, Hyo-Sang Shin, and Antonios Tsourdos

***Abstract*—This paper considers the problem of two-dimensional constrained trajectory optimization of a point-mass aerial robot for constant-maneuvering target localization using bearing-only measurement. A performance metric that can be utilized in trajectory optimization to maximize target observability is proposed first based on geometric conditions. One-step optimal maneuver that maximizes the observability criterion is then derived analytically for moving targets. The heading angle constraint is also incorporated in the proposed optimal maneuver derivation to support practical application. Numerical simulations with some comparisons are presented to validate the analytical findings.**

***Index Terms*—Bearing-only measurement, physical constraint, target localization, trajectory optimization.**

I. Introduction

SMALL-SCALE robots, especially in aerial robotics, have been widely used and show great potentials in both civilian and military applications. The issue is that the operations of these small-scale aerial robots are constrained by limited payload, power, and endurance. This has led to proliferation of numerous lightweight, low-cost, and energy efficient onboard sensors. A particularly interesting mission for small robots is to localize and track targets of interest [1]–[6]. However, since most low-cost and lightweight sensors are passive sensors, e.g., optical or infrared camera, the utilization of these sensors in target localization usually suffers from the observability problem. Therefore, system observability becomes a fundamental requirement for bearing-only target localization [7].

When employing passive sensors for target localization, only the bearing or line-of-sight (LOS) angle can be measured. It is well-known that the bearing angle measurement cannot provide relative range information without any relative maneuver, that is, the relative states between the observer and the target are unobservable under certain conditions. Fundamentally, only when the target is observable, the observer can localize or track the target. To this end, observability conditions for bearing-only target localization are widely studied over past years. Nardone and Aidala [7] and Hammel and Aidala [8] derived closed-form observability criteria for two-dimensional (2-D) constant moving target localization. The results revealed that the observer acceleration should satisfy a certain condition to guarantee a certain degree of observability, which seem to be the pioneer works in this domain. As an extension of the work presented in [7], Tahk *et al.* [9] derived the observability condition for constant accelerating target localization. The work presented in [10] further generalized the condition to Nth order target dynamics and revealed that the observability condition derived in [7] is only necessary but not sufficient. Woffinden [11] and Geller [12] proposed a general observability condition for angle-only navigation to determine the relative orbital motion and provided the conditions where the unobservable maneuvers occur. By using a pseudolinear structure of the pseudomeasurements, necessary and sufficient conditions, that can be applied to general target motions, for observability analysis were proposed in [13]. With a few assumptions, a closed-form metric that could be utilized to quantify system observability for maneuvering target was derived in [14]. This analytical criterion essentially coincides with previously suggested metric given in [15] for constant moving targets.

Since system observability depends on the relative geometry [16]–[19], the trajectory of the observer poses a direct impact on the achievable estimation performance [20]–[23]. For this reason, it is known that trajectory optimization, maximizing system observability, can yield significant benefits for target tracking with bearing-only measurement to improve the quality of perceptual results [24]. Optimal observer trajectory that maximizes system observability is generated in [3], [21], and [25]–[27] by maximizing the determinant of Fisher information matrix (FIM) over a finite time horizon. The rationale of using FIM for a cost function, also known as objective function, lies in that the inverse of FIM prescribes a lower bound of the estimation error covariance of an unbiased filter [28]. To minimize the sensing time and ensure bounded estimation uncertainty, an approximate algorithm to plan the bearing data gathering path was proposed in [29], where the determinant of FIM was utilized to quantify the estimation performance. Different from [25] and [26], the determinant of error covariance matrix was employed as the cost function in [30] and [31] for trajectory optimization. Using the same objective function, exhaustive search was utilized in [32] to find optimal sensor heading commands with motion constraints. On the basis of the nonlinear observability analysis tool, i.e., Observability Gramian (OG) [33], Salaris *et al.* [34] and Cognetti *et al.* [35] provided online gradient

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descent path planning strategies to actively maximize the smallest eigenvalue of the OG. A cautious greedy active sensing strategy was proposed in [36] and [37] for localizing a stationary target with bearing-only measurement in minimum time. Taking communication range and no-fly zone constraints into account, a numerical trajectory planning for target localization with bearing-only measurement using multiple robots was proposed in [38]. An algorithm for cooperative active localization of stationary targets using mobile bearing sensors was suggested in [39] by leveraging the largest eigenvalue of FIM as the cost function. To improve bearing-only observability while simultaneously considering other mission objectives, multiobjective trajectory optimization problem was formulated and solved in [21], [40], and [41]. Unfortunately, this particular problem formulated is not amendable to simple analytical solution. Hence, these approaches use numerical methods to find optimal solutions and consequently require high onboard computational power. As the onboard computational power is limited in low-cost and lightweight robots, those approaches are not suitable for such robots. Moreover, optimization over a finite time horizon requires accurate target dynamics, which becomes prohibitive in practice, especially for hostile targets. These issues can be alleviated by applying one-step analytical optimization as in [42]: optimal trajectory was obtained for ground robots by minimizing the trace of the estimation error covariance matrix in one step in the paper. However, since physical constraints of the robot are not rigorously considered, the solution derived in [42] might become infeasible for the robot.

This paper aims to develop an analytically optimal trajectory optimization algorithm in consideration of physical constraints for constant-maneuvering target localization with bearing-only measurement. The contributions of this paper are twofold.

1) Unlike previous approaches using FIM or error covariance matrix, this paper finds a performance measure of observability based on geometric conditions and utilizes it as a pertinent cost function. While it is known that the relative geometry conditions determine the observability, there has been no attempt to obtain a geometry-based estimation performance measure for bearing-only-based target localization available in the literature. Determining the performance measure, we provide a geometric interpretation on how the relative geometry influences the system observability. Such interpretation enables more intuitive and perhaps better insights to the relationship between the geometric condition and observability.
2) This paper derives a closed-form solution that maximizes the geometric observability measure formulated and meets physical constraints such as minimum turning rate. The resultant analytical solution, given as heading angle input command, is simple to be implemented in practice. Up to the best of our knowledge, no closed-form solution with a physical constraint is available in existing literature.

Realistic scenarios are simulated to illustrate and evaluate the performance of the proposed algorithm, and the results are compared with those of the existing analytical approach proposed in [42] and the approach in [14]. The simulation results reveal that the proposed algorithm meets the considered physical constraint unlike the approach from [42]. They also show that optimizing the proposed geometric performance metric generates higher system observability than optimizing the analytical measure derived in [14], thus leading to improved estimation performance.

This paper is organized as follows. Section II presents some preliminaries and backgrounds. Section III derives a geometric metric for observability analysis, followed by the optimal maneuver derivation shown in Section IV. Then, some simulations and conclusions are offered.

II. Backgrounds and Preliminaries

This section provides some necessary preliminaries of vehicle kinematics model and bearing-only measurement model to facilitate the analysis in the following sections.

A. Vehicle Kinematics

This paper assumes that the observer is equipped with a high-performance low-level flight control system that provides velocity tracking, heading, and altitude hold functions. This paper aims to design guidance input to this low-level controller for target localization and only concerns the 2-D motions. The rationale of leveraging simple 2-D engagement is that typical aerial robots provide roll stabilization and therefore the three-dimensional (3-D) guidance problem can be decoupled into horizontal and vertical channels using the well-known separation concept. The vehicle's kinematics in a 2-D environment is given by

$$\begin{aligned} \dot{x}_o &= V_o \cos\gamma_o \\ \dot{y}_o &= V_o \sin\gamma_o \end{aligned} \tag{1}$$

where (x_o, y_o) stands for the observer's position in an inertial coordinate. γ_o is the observer's heading angle and $V_o \in (0, V_{o,\max}]$ denotes the observer's velocity with $V_{o,\max}$ being the maximum permissible velocity. Note that the observer's velocity V_o is assumed to be larger than the target's velocity in this paper.

In practice, the observer heading change between two consecutive time steps is constrained due to physical turning rate limitation as

$$|\gamma_{o,k} - \gamma_{o,k-1}| \le \gamma_{\max} \triangleq \omega_{\max} T_s \tag{2}$$

where $\gamma_{o,k}$ represents the heading angle at time step k, $\omega_{\max}$ the maximum permissible turning rate of the observer, and T_s the sampling time.

B. Measurement Model

Fig. 1 shows the geometric relationship between the observer and the target at time step k. $X-O-Y$ stands for the inertial coordinate. The notations $\mathbf{r}_{o,k} = [x_{o,k}, y_{o,k}]^T$ and $\mathbf{r}_{t,k} = [x_{t,k}, y_{t,k}]^T$ represent the position vectors of the observer and the target at time step k, respectively. $\mathbf{r}_k = \mathbf{r}_{o,k} - \mathbf{r}_{t,k}$ denotes the relative position vector between the observer and the target. With heading constraint (2), the maximum region the observer can travel at current time step is described by a sector, as shown in Fig. 1. At time step k, the observer only has access to

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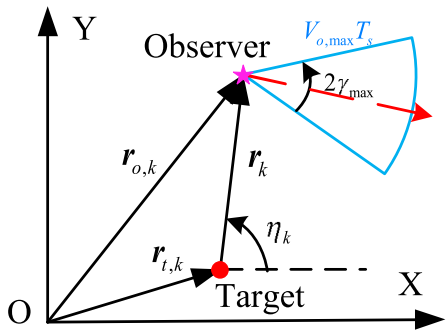


Fig. 1. Geometric relationship between the observer and the target at time step k in an inertial coordinate. The observer is represented by a magenta pentagram and the red circle denotes the target. The red vector stands for the observer heading direction at previous time step. The maximum permissible move region of the observer at current time instant is given by the blue sector with radius $V_{o,\max}T_s$ and angle $2\gamma_{\max}$.

its bearing angle measurement η_k toward the target. Therefore, the sensor measurement z_k is given by

$$z_k = \eta_k + \varepsilon \tag{3}$$

where ε denotes the sensor measurement noise.

Let $\mathbf{r}_k = [x_k, y_k]^T$, then, the sensor measurement z_k can also be formulated as

$$z_k = \arctan\left(\frac{y_k}{x_k}\right) + \varepsilon. \tag{4}$$

From (4), it is clear that the relative distance is not directly measurable by bearing angle η_k, that is, (x_k, y_k) and $(\mu x_k, \mu y_k)$ with $\mu \in \mathbb{R}^+$ provide the same bearing angle η_k. Therefore, certain maneuvers have to be performed to make the range observable in a certain degree.

Remark 1: Note that the bearing angle in (4) is the direction of $\mathbf{r}_k$ with respect to a world/inertial frame. In practice, however, observers with onboard sensors, i.e., cameras, can only provide the direction of $\mathbf{r}_k$ with respect to the body frame. Therefore, additional observer attitude information from inertial measurement unit (IMU) is required to obtain η_k. For simplicity, we assume that the IMU provides accurate attitude information by a well-tuned filter and therefore only sensor noise is considered in measurement model (4). Note that this assumption has been widely used in active target localization with bearing-only measurement [14], [39], [42], [43]. Although the IMU output has inevitable noise, it can be accumulated in ε and therefore the algorithm developed is still valid.

III. Geometric Analysis and Problem Formulation

This section first derives a geometric metric that can be utilized in trajectory optimization to maximize system observability. Then, the problem formulation of the paper is stated.

A. Geometric Metric for Observability Maximization

Change in the vehicle's velocity and its direction over a short interval, like over T_s, is usually negligible. Hence, for simplicity, it is assumed that both the observer and the target are piecewise nonmaneuvering, i.e., moving with constant speed and constant direction between two consecutive time instants. Fig. 2 shows the geometric relationship between the observer and the target in two consecutive time instants, where Fig. 2(a) shows the geometric relationship in an inertial frame and Fig. 2(b) provides the geometric relationship in a relative frame. Notice that the relative frame can accommodate moving target since the relative position vector $\mathbf{r}_k$ directly contains the information on target movement. In Fig. 2, $\delta\mathbf{r}_k$ represents the relative maneuver at time step k. σ is the separation angle between two consecutive relative position vectors.

Between two consecutive time instants, it is clear from Fig. 2 that the relative position vector $\mathbf{r}_{k+1}$ at next time step can be uniquely determined by triangulation if $\sigma \neq 0$ and $\mathbf{r}_k$ is available to the observer. Based on this observation, we assume that the relative position vector $\mathbf{r}_k$ is known for finding the optimal observer maneuver vector at time instant k in one-step trajectory optimization. The validity of this assumption will be proved through a detailed observability analysis shown in Section IV-E. Note that, if $\sigma = 0$, $\mathbf{r}_{k+1} = k_1\mathbf{r}_k$, $k_1 \in \mathbb{R}^+$. Then, it is impossible to distinguish the next step relative position vector from the current step relative position vector regardless of the angle measurement. Hence, the relative maneuver should satisfy the condition $\sigma \neq 0$, which coincides with the necessary observability condition that the LOS rate $\dot{\eta} = \sigma/T_s$ cannot be zero, derived in [9] for a nonmaneuvering target. Therefore, under the assumption that $\mathbf{r}_k$ is available, the robot can localize the target only when the robot-target relative geometry satisfies the condition $\sigma \neq 0$.

Let $\delta\rho \triangleq CE$ be the relative position estimation error at time step $k+1$ caused by the sensor bearing measurement error ε. Based on the law of sines, we have

$$\triangle OCE\text{: } \frac{\delta\rho}{\sin\varepsilon} = \frac{OE}{\sin\left[\pi - (\sigma + \varepsilon/2)\right]} = \frac{OE}{\sin\left(\sigma + \varepsilon/2\right)} \tag{5}$$

$$\triangle ODE\text{: } \frac{\|\delta\mathbf{r}_k\|}{\sin\left(\sigma - \varepsilon/2\right)} = \frac{OE}{\sin\alpha} \tag{6}$$

$$\triangle OAB\text{: } \frac{\|\mathbf{r}_{k+1}\|}{\sin\alpha} = \frac{\|\delta\mathbf{r}_k\|}{\sin\sigma}. \tag{7}$$

Combining (5) and (6), one can imply that

$$\delta\rho = \|\delta\mathbf{r}_k\|\left(\frac{\sin\alpha\sin\varepsilon}{\sin^2\sigma - \sin^2\left(\varepsilon/2\right)}\right) \tag{8}$$

where the fact that $\sin\left(a+b\right)\sin\left(a-b\right) = \sin^2 a - \sin^2 b$ is utilized in the derivation.

Assume that the sensor bearing measurement error ε is small, (8) can be reduced to

$$\delta\rho = \varepsilon\|\delta\mathbf{r}_k\|\left(\frac{\sin\alpha}{\sin^2\sigma}\right). \tag{9}$$

Substituting (7) into (9) yields the error square as

$$\delta\rho^2 = \varepsilon^2\left(\frac{\|\mathbf{r}_{k+1}\|}{\sin\sigma}\right)^2 \tag{10}$$

which reveals that the relative position estimation (or target localization) error at next time step is characterized by the sensor measurement error ε, the separation angle σ, and the relative

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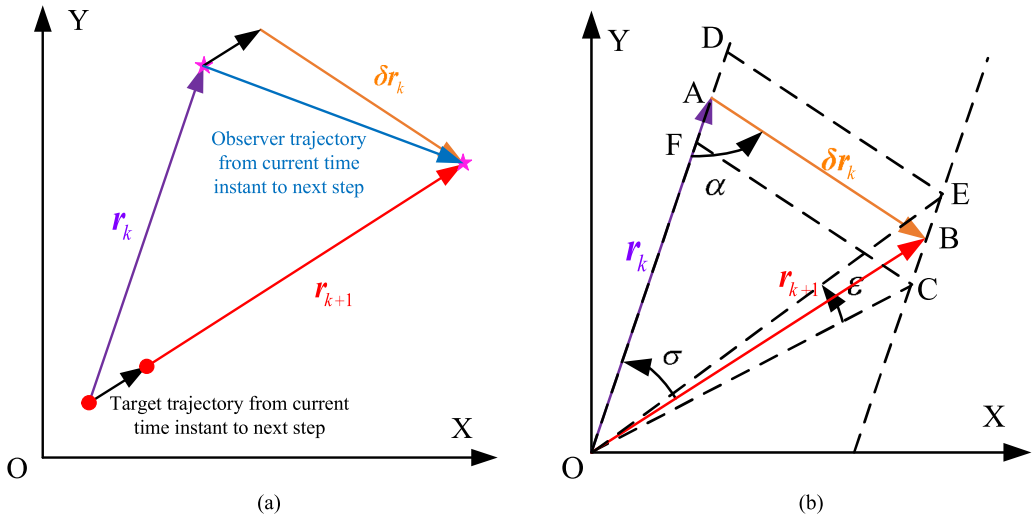


Fig. 2. Geometric relationship between the observer and the target in two consecutive time instants. (a) Geometric relationship in an inertial frame. The black line represents the target movement trajectory and the blue line stands for the observer movement trajectory at current time instant. The orange line refers to the relative movement trajectory. (b) Geometric relationship in a relative frame. The relative frame in (b) is defined by using the relative motion kinematics, e.g., relative range $\mathbf{r}_k$, and attaching the origin at the target position at time instant k. In (b), line $\overline{CE}$ is parallel to line $\overline{FD}$. Both lines $\overline{DE}$ and $\overline{FC}$ are parallel to line $\overline{AB}$ in (b).

range $\|\mathbf{r}_{k+1}\|$. This provides us better insights to the influence of relative geometric configuration on system observability.

Remark 2: Notice that (10) closely resembles the expression of geometric dilution of precision, which is commonly leveraged in satellite navigation and geomatics engineering: the term $(\|\mathbf{r}_{k+1}\|/\sin\sigma)^2$ determines the sensitivity of estimation performance to bearing measurement errors.

B. Problem Formulation

The aim of the trajectory optimization is to determine optimal maneuvers that minimize target localization error and also respect the physical constraints given in (2). To accomplish the aim, this paper formulates a discrete-time trajectory optimization problem, which is denoted as CTO_1.

CTO_1: find the observer maneuver at time step k, $\mathbf{v}_{o,k}$, which maximizes the following objective function J:

$$J = \left(\frac{\sin\sigma}{\|\mathbf{r}_{k+1}\|}\right)^2 \tag{11}$$

subject to

$$\begin{aligned} &|\gamma_{o,k} - \gamma_{o,k-1}| \le \omega_{\max} T_s \\ &0 < V_{o,k} \le V_{o,\max}. \end{aligned} \tag{12}$$

Note that the observer maneuver at time step k is given as

$$\mathbf{v}_{o,k} = [V_{o,k} T_s \cos\gamma_{o,k}, V_{o,k} T_s \sin\gamma_{o,k}]^T. \tag{13}$$

The cost function J is determined in consideration of $\delta\rho^2$ obtained in (10): since ε is determined by the sensing device, a practical way to reduce the estimation error is to find an optimal one-step observer maneuver that maximizes the cost function J given by (11).

Remark 3: Geometrically, the minimization of $\|\mathbf{r}_{k+1}\|$ and the maximization of $|\sin\sigma|$ are two conflict objectives. Therefore, the information maximizing maneuver obtained from $\max J$ will provide a proper balance between forcing σ to approach $\pm\pi/2$ and reducing the relative range. This means that the resulting optimal solution satisfies $-\pi/2 < \sigma < \pi/2$.

Remark 4: Note that, since (11) is provided in a closed form, it permits simple weighed inclusion inside a global trajectory optimization problem as an auxiliary objective, allowing for an optimal solver to find maneuvers that are beneficial in terms of observability enhancement while considering other objectives.

Remark 5: Assume that the range variation between two successive instants is negligible, i.e., $\|\mathbf{r}_k\| \approx \|\mathbf{r}_{k+1}\|$ and the separation angle is small, i.e., $\sin(\sigma) \approx \sigma$, Anjaly and Ratnoo [14] revealed that the determinant of the incremental of FIM at time step k is lower bounded by

$$|\Delta F_k| = |F_{k+1} - F_k| \ge c\left(\frac{\sigma}{\|\mathbf{r}_k\|}\right)^2 \tag{14}$$

where c is a constant depending on the measurement noise statistics and the time step. The notation F_k denotes the FIM at time instant k. It follows from (14) that maximizing $(\sigma/\|\mathbf{r}_k\|)^2$ can increase the lower bound of $|\Delta F_k|$, leading to the increase of observability between two consecutive time steps. Therefore, $\hat{J} = (\sigma/\|\mathbf{r}_k\|)^2$ was utilized as the cost function in [14] to find optimal observer trajectory for information maximization.

One can note that both J and $\hat{J}$ share a similar characteristic that target observability can be enhanced by increasing the absolute value of the separation angle σ. Apart from this similar property, the difference between cost functions J and $\hat{J}$ is clear: $\hat{J}$ ignores the variation of $\|\mathbf{r}_k\|$ while the proposed geometric measure relaxes this assumption. Conceptually, one can geometrically conclude from Fig. 2(b) that smaller $\|\mathbf{r}_k\|$ provides lower estimation error with same bearing noise ε. With this in mind, it is expected that maximizing J can generate improved estimation performance, compared to maximizing $\hat{J}$. It is

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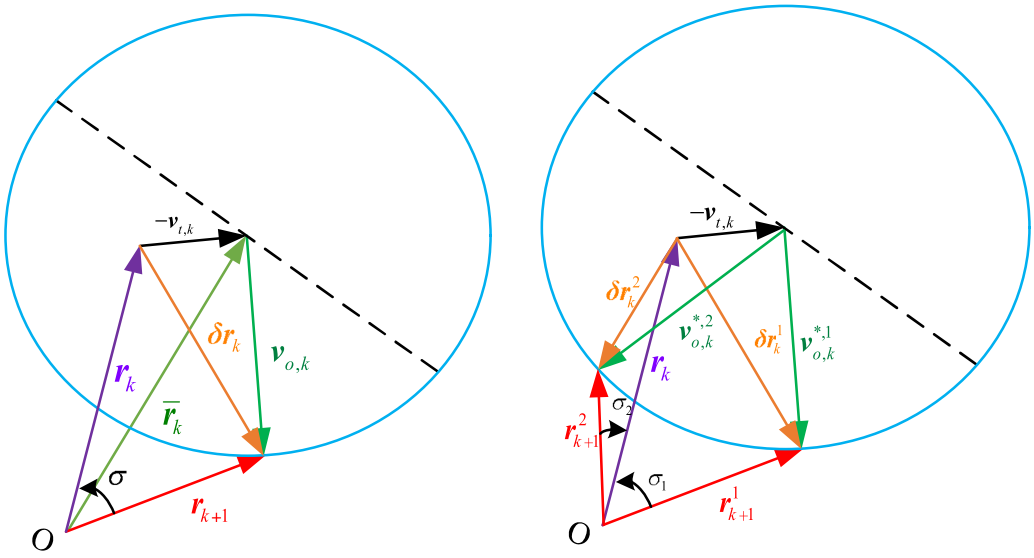


Fig. 3. (a) Geometric illustration for moving target localization without turning rate limit in the relative frame. The blue circle determines the maximum permissible region that the observer can travel at current time step. $\bar{\mathbf{r}}_k = \mathbf{r}_k - \mathbf{v}_{t,k}$ is an auxiliary vector utilized in the analysis. (b) Geometric illustration of candidate optimal heading solutions for moving target localization in the relative frame.

worthy to point out that the proposed cost function J gradually converges to $\hat{J}$ when $T_s \to 0$.

Remark 6: Different from [14], Zhou and Roumeliotis [42] provided a solution of trajectory optimization by exploiting the trace of error covariance as a cost function, which directly characterizes the average estimation variance. However, the turning rate limit constraint was not considered in the optimization, which means that the solution obtained in [42] might become infeasible in practice.

Remark 7: It should be pointed out that both [14] and [42] formulated their cost functions directly from classical estimation theory. However, the proposed cost function J is derived using simple geometric condition with clear physical meanings. This obviously reveals that how the observer-target relative geometry influences the estimation accuracy.

IV. Derivation of Optimal Maneuver for Target Localization

This section will propose an analytical optimal maneuver that maximizes the cost function J for target localization with bearing-only measurement. We first derive the optimal solution without heading constraint and then extend to the case that the robot has limited turning rate to change its heading angle. Finally, target observability under the proposed approach will be analyzed.

A. Optimal Maneuver Without Heading Constraints

Excluding the heading constraint, CTO_1 reduces to CTO_2.

CTO_2: find the observer maneuver at time step k, $\mathbf{v}_{o,k}$, which maximizes the following objective function J:

$$J = \left(\frac{\sin\sigma}{\|\mathbf{r}_{k+1}\|} \right)^2 \tag{15}$$

subject to

$$0 < V_{o,k} \le V_{o,\max}. \tag{16}$$

Let $\mathbf{v}_{t,k} = [v_{tx,k}, v_{ty,k}]^T \triangleq [V_{t,k}T_s \cos\gamma_{t,k}, V_{t,k}T_s \sin\gamma_{t,k}]^T$ be the target maneuver vector at time step k. Since the target velocity vector at current time step can be obtained from Kalman filter, $\mathbf{v}_{t,k}$ is known in trajectory optimization. Fig. 3(a) shows the geometric relationship between the observer and the target within two consecutive time steps. Since $\mathbf{v}_{t,k}$ is fixed, $\bar{\mathbf{r}}_k$ is known in trajectory optimization. The analytical solution of CTO_2 can be obtained using Lemmas 1–3.

Lemma 1: Given the observer velocity $V_{o,k}$, the candidate optimal heading angle at time step k without any constraint is given by

$$\gamma_{o,k}^{*,1} = \arcsin\left(-\frac{2V_{o,k}T_s \mathbf{r}_k^T \cdot \bar{\mathbf{r}}_k}{\sqrt{a^2+b^2}} \right) - \vartheta \tag{17}$$

$$\gamma_{o,k}^{*,2} = \pi - \arcsin\left(-\frac{2V_{o,k}T_s \mathbf{r}_k^T \cdot \bar{\mathbf{r}}_k}{\sqrt{a^2+b^2}} \right) - \vartheta \tag{18}$$

with $\sin\vartheta = b/\sqrt{a^2+b^2}$ and $\cos\vartheta = a/\sqrt{a^2+b^2}$, where

$$a = \|\mathbf{r}_k\|^2 (y_k - 2v_{ty,k}) + 2x_k v_{tx,k} v_{ty,k} + y_k \left[v_{ty,k}^2 - v_{tx,k}^2 + (V_{o,k}T_s)^2 \right] \tag{19}$$

$$b = \|\mathbf{r}_k\|^2 (x_k - 2v_{tx,k}) + 2y_k v_{tx,k} v_{ty,k} + x_k \left[v_{tx,k}^2 - v_{ty,k}^2 + (V_{o,k}T_s)^2 \right]. \tag{20}$$

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Proof: For moving target, the relative maneuver vector at time step k can be obtained as

$$\begin{aligned}\delta\mathbf{r}_k &= \mathbf{v}_{o,k} - \mathbf{v}_{t,k} \\ &= \left[V_{o,k}T_s\cos\gamma_{o,k} - v_{tx,k}, V_{o,k}T_s\sin\gamma_{o,k} - v_{ty,k}\right]^T.\end{aligned} \tag{21}$$

Then, the relative position vector at time step $k+1$ is given by

$$\begin{aligned}\mathbf{r}_{k+1} &= \mathbf{r}_k + \delta\mathbf{r}_k \\ &= \big[x_k + V_{o,k}T_s\cos\gamma_{o,k} - v_{tx,k}, y_k \\ &\quad + V_{o,k}T_s\sin\gamma_{o,k} - v_{ty,k}\big]^T.\end{aligned} \tag{22}$$

As depicted in Fig. 2(b), the separation angle σ between two consecutive relative position vectors is determined by

$$\sigma = \arccos\left(\frac{\mathbf{r}_k^T\cdot\mathbf{r}_{k+1}}{\|\mathbf{r}_k\|\,\|\mathbf{r}_{k+1}\|}\right). \tag{23}$$

Substituting (23) into (11) gives

$$J = \frac{1-\cos^2\sigma}{x_{k+1}^2+y_{k+1}^2} = \frac{\left(x_k y_{k+1} - x_{k+1}y_k\right)^2}{\left(x_k^2+y_k^2\right)\left(x_{k+1}^2+y_{k+1}^2\right)^2}. \tag{24}$$

The partial derivative of J with respect to $\gamma_{o,k}$ provides a constraint for identifying the optimal heading as

$$\frac{\partial J}{\partial\gamma_{o,k}} = 0. \tag{25}$$

Substituting (22) into (24) and after some tedious but simple algebraic manipulations, (25) reduces to (26) as shown at the bottom of this page.

From (26), the solution of $\partial J/\partial\gamma_{o,k} = 0$ can be determined from

$$\begin{aligned}&x_k\left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right) - y_k\big(x_k - v_{tx,k} \\ &\quad + V_{o,k}T_s\cos\gamma_{o,k}\big) = 0\end{aligned} \tag{27}$$

Note that, if $x_k\left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right) - y_k(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}) = 0$, we have $\mathbf{r}_{k+1} = k_2\mathbf{r}_k,\ k_2\in\mathbb{R}$. This provides zero observability of the target by the angle measurement, which minimizes the cost function J. Therefore, this should be excluded from the solution candidates. Together with the fact that the cost function J is continuous, the candidate optimal heading solutions can be obtained from condition (28) as shown at the bottom of this page. Further simplifying (28) yields

$$a\sin\gamma_{o,k} + b\cos\gamma_{o,k} = -2V_{o,k}T_s\mathbf{r}_k^T\cdot\bar{\mathbf{r}}_k. \tag{29}$$

Note that

$$a\sin\gamma_{o,k} + b\cos\gamma_{o,k} = \sqrt{a^2+b^2}\sin\left(\gamma_{o,k}+\vartheta\right). \tag{30}$$

Since both a and b are known, the candidate optimal observer heading angle $\gamma_{o,k}^*$ can be directly obtained from (29) and (30) as

$$\gamma_{o,k}^{*,1} = \arcsin\left(-\frac{2V_{o,k}T_s\mathbf{r}_k^T\cdot\bar{\mathbf{r}}_k}{\sqrt{a^2+b^2}}\right) - \vartheta \tag{31}$$

$$\gamma_{o,k}^{*,2} = \pi - \arcsin\left(-\frac{2V_{o,k}T_s\mathbf{r}_k^T\cdot\bar{\mathbf{r}}_k}{\sqrt{a^2+b^2}}\right) - \vartheta. \tag{32}$$

■

Remark 8: It follows from (31) and (32) that the candidate optimal heading solution with a specific observer velocity $V_{o,k}$ depends on current relative position as well as target's velocity magnitude and its corresponding direction. The geometric illustration of optimal observer maneuvers for moving target is shown in Fig. 3(b), where $-\mathbf{v}_{t,k}$ is assumed to be located on the right-hand side (RHS) of $\mathbf{r}_k$. Similar geometry can also be obtained if $-\mathbf{v}_{t,k}$ is located on the left-hand side (LHS) of $\mathbf{r}_k$. From Fig. 3(b), we can note that both $\gamma_{o,k}^{*,1}$ and $\gamma_{o,k}^{*,2}$ try to reduce the relative range between the observer and the target. In the meantime, these two solutions try to increase the absolute value of the separation angle σ to improve target observability.

Given the robot velocity $V_{o,k}$, Lemma 1 indicates that there exist two candidate heading angle solutions that locally maximize the cost function J, i.e., these two solutions are local maxima. The following lemma analyzes the location of these two solutions. Without loss of generality, we assume that $-\mathbf{v}_{t,k}$

$$\begin{aligned}\frac{\partial J}{\partial\gamma_{o,k}} &= 2V_{o,k}T_s\left[\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)^2 + \left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right)^2\right] \\ &\quad\times\frac{x_k\left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right) - y_k\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)}{\left[\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)^2 + \left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right)^2\right]^4} \\ &\quad\times\Big\{\left(x_k\cos\gamma_{o,k} + y_k\sin\gamma_{o,k}\right)\left[\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)^2 + \left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right)^2\right] \\ &\quad + 2\left[x_k\left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right) - y_k\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)\right] \\ &\quad\times\left[\sin\gamma_{o,k}\left(x_k - v_{tx,k}\right) - \cos\gamma_{o,k}\left(y_k - v_{ty,k}\right)\right]\Big\}\end{aligned} \tag{26}$$

$$\begin{aligned}&\left(x_k\cos\gamma_{o,k} + y_k\sin\gamma_{o,k}\right)\left[\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)^2 + \left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right)^2\right] \\ &\quad + 2\left[x_k\left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right) - y_k\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)\right] \\ &\quad\times\left[\sin\gamma_{o,k}\left(x_k - v_{tx,k}\right) - \cos\gamma_{o,k}\left(y_k - v_{ty,k}\right)\right] = 0.\end{aligned} \tag{28}$$

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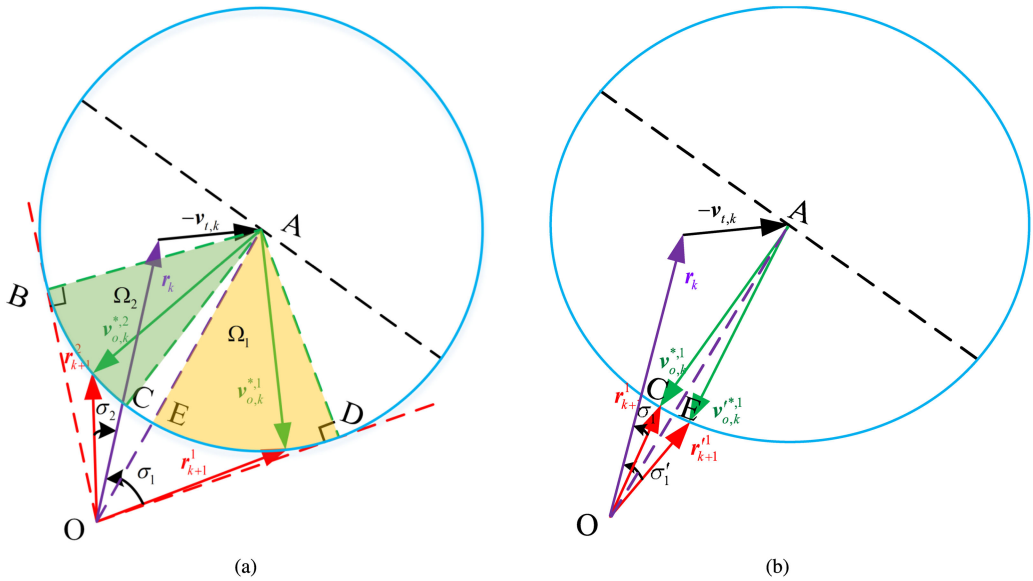


Fig. 4. Geometric illustration of optimal maneuver location in the relative frame. (a) One candidate heading solution $\mathbf{v}^{*,1}_{o,k}$ locates inside sector ADE excluding line $\overline{AD}$ (Ω_1 in the figure) and another candidate heading solution $\mathbf{v}^{*,2}_{o,k}$ locates inside sector ABC excluding lines $\overline{AB}$, $\overline{AC}$ (Ω_2 in the figure). (b) Proof of the fact that $\mathbf{v}^{*,1}_{o,k} \notin$ sector ACE.

is located on the RHS of $\mathbf{r}_k$, as shown in Fig. 4, in the following analysis. Similar results can also be obtained when $-\mathbf{v}_{t,k}$ is located on the LHS of $\mathbf{r}_k$.

Lemma 2: Let Ω_1 be sector ADE that excludes line $\overline{AD}$ and Ω_2 be sector ABC that excludes lines $\overline{AB}$, $\overline{AC}$ [refer to Fig. 4(a)]. Then, the heading solution given by Lemma 1 satisfies $\mathbf{v}^{*,1}_{o,k} \in \Omega_1$ and $\mathbf{v}^{*,2}_{o,k} \in \Omega_2$.

Proof: The proof of Lemma 2 is given in two steps. The first step proves that the observer maneuver vector solutions, given by Lemma 1, are located inside sector ABD. The second step shows that $\mathbf{v}^{*,1}_{o,k} \in \Omega_1$ and $\mathbf{v}^{*,2}_{o,k} \in \Omega_2$.

Step 1: As discussed earlier, the maximization of J provides a balance between the minimization of $\|\mathbf{r}_{k+1}\|$ and the maximization of $|\sin\sigma|$. If the observer maneuver vector $\mathbf{v}_{o,k}$ coincides with $\overrightarrow{AC}$, one can imply that $\mathbf{r}_{k+1} = \lambda \mathbf{r}_k$ with $\lambda \in \mathbb{R}^+$. This means that both $\mathbf{r}^2_{k+1}$ and $\sin^2\sigma$ achieve their minimum values. Therefore, the proof of Step 1 is equivalent to the proof of the fact that the optimal observer maneuver vector that maximizes $\sin^2\sigma$ is perpendicular to next step relative position vector $\mathbf{r}_{k+1}$.

Consider the cost function $\bar{J} = \sin^2\sigma$. Evaluating the partial derivative of $\bar{J}$ with respect to $\gamma_{o,k}$ gives in (33) as shown at the bottom of this page.

Solving $\partial\bar{J}/\partial\gamma_{o,k} = 0$ gives in (34) and (35) as shown at the bottom of this page.

Note that, if $x_k(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}) - y_k(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}) = 0$, we have $\mathbf{r}_{k+1} = k_2\mathbf{r}_k$, $k_2 \in \mathbb{R}$. This will minimize the cost function $\bar{J}$ since $\sigma = 0$. Therefore,

$$\begin{aligned}\frac{\partial\bar{J}}{\partial\gamma_{o,k}} = {} & \frac{2V_{o,k}T_s\left[x_k\left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right) - y_k\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)\right]}{\left[\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)^2 + \left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right)^2\right]^2} \\ & \times\Big\{\left(x_k\cos\gamma_{o,k} + y_k\sin\gamma_{o,k}\right)\left[\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)^2 + \left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right)^2\right] \\ & + \left[x_k\left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right) - y_k\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)\right] \\ & \times\left[\sin\gamma_{o,k}\left(x_k - v_{tx,k}\right) - \cos\gamma_{o,k}\left(y_k - v_{ty,k}\right)\right]\Big\}\end{aligned} \tag{33}$$

$$x_k\left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right) - y_k\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right) = 0. \tag{34}$$

$$\begin{aligned}&\left(x_k\cos\gamma_{o,k} + y_k\sin\gamma_{o,k}\right)\left[\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)^2 + \left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right)^2\right] \\ &\quad + \left[x_k\left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right) - y_k\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right)\right] \\ &\quad \times\left[\sin\gamma_{o,k}\left(x_k - v_{tx,k}\right) - \cos\gamma_{o,k}\left(y_k - v_{ty,k}\right)\right] = 0\end{aligned} \tag{35}$$

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the candidate optimal heading satisfies condition (35). Simplifying (35) yields

$$\begin{aligned}&[(x_k - v_{tx,k})\cos\gamma_{o,k} + (y_k - v_{ty,k})\sin\gamma_{o,k} + V_{o,k}T_s]\\ &\quad\times\left(x_k^2+y_k^2-x_k v_{tx,k}-y_k v_{ty,k}+x_k V_{o,k}T_s\cos\gamma_{o,k}\right.\\ &\quad\left.+\ y_k V_{o,k}T_s\sin\gamma_{o,k}\right)=0\end{aligned}\tag{36}$$

which is equivalent to

$$\left(\mathbf{r}_{k+1}^T\cdot\mathbf{v}_{o,k}\right)\left(\mathbf{r}_k^T\cdot\mathbf{r}_{k+1}\right)=0.\tag{37}$$

Since the optimal maneuver satisfies the condition that $\mathbf{r}_k^T\cdot\mathbf{r}_{k+1}\neq 0$, the final solution that maximizes $\bar{J}$ is given by $\mathbf{r}_{k+1}^T\cdot\mathbf{v}_{o,k}=0$. This clearly indicates that the optimal observer maneuver vector that maximizes $\sin^2\sigma$ is perpendicular to next step relative position vector $\mathbf{r}_{k+1}$.

Step 2: Note that if $\mathbf{v}_{o,k}=\overrightarrow{AC}$, the cost function J achieves the minimum value as $J(\overrightarrow{AC})=0$. Therefore, the cost function J monotonically increases when the observer maneuver vector $\mathbf{v}_{o,k}$ rotates from $\overrightarrow{AC}$ to $\mathbf{v}_{o,k}^{*,1}$ or to $\mathbf{v}_{o,k}^{*,2}$ since there is no solution $\partial J/\partial\gamma_{o,k}=0$ that is located between $\mathbf{v}_{o,k}^{*,1}$ and $\overrightarrow{AC}$ or between $\mathbf{v}_{o,k}^{*,2}$ and $\overrightarrow{AC}$. This implies that $\mathbf{v}_{o,k}^{*,2}\in\Omega_2$ and $\mathbf{v}_{o,k}^{*,1}$ locates in sector ACD that excludes lines $\overline{AC}$, $\overline{AD}$. Now, assume that $\mathbf{v}_{o,k}^{*,1}$ locates inside sector ACE. Then, according to symmetric property, there always exists a vector $\mathbf{v}'^{*,1}_{o,k}$ such that $\left\|\mathbf{r}'^1_{k+1}\right\|=\left\|\mathbf{r}^1_{k+1}\right\|$, as shown in Fig. 4(b). This means that

$$\begin{aligned}J\left(\mathbf{v}'^{*,1}_{o,k}\right)&=\left(\frac{\sin\sigma_1'}{\left\|\mathbf{r}'^1_{k+1}\right\|}\right)^2=\left(\frac{\sin\sigma_1'}{\left\|\mathbf{r}^1_{k+1}\right\|}\right)^2\\&>\left(\frac{\sin\sigma_1}{\left\|\mathbf{r}^1_{k+1}\right\|}\right)^2=J\left(\mathbf{v}^{*,1}_{o,k}\right)\end{aligned}\tag{38}$$

which reveals that $\mathbf{v}_{o,k}^{*,1}\notin$ sector ACE. Therefore, $\mathbf{v}_{o,k}^{*,1}\in\Omega_1$.■

From Lemma 1, it is clear that the candidate optimal heading command solutions depend on observer's velocity. Now, let us analyze the effect of observer's velocity $V_{o,k}$ on cost function J.

Lemma 3: The cost function J monotonically increases with respect to the observer's velocity $V_{o,k}$ when $\mathbf{v}_{o,k}\in\Omega_1\cup\Omega_2$.

Proof: Evaluating the partial derivative of J with respect to $V_{o,k}$ gives in (39) as shown at the bottom of this page.

Let $\Omega_1=\Omega_3\cup\Omega_4$, where Ω_3 be sector ADF that excludes line $\overline{AD}$ and Ω_4 be sector AEF that excludes line $\overline{AF}$ [shown in Fig. 5(a), where line $\overline{AF}$ is parallel to $\mathbf{r}_k$]. The following three different conditions are considered.

Condition 1: When $\mathbf{v}_{o,k}\in\Omega_2$, which means that $\mathbf{v}_{o,k}$ is located on the LHS of $\mathbf{r}_k$, we have

$$y_k\cos\gamma_{o,k}-x_k\sin\gamma_{o,k}<0\tag{40}$$

and $\mathbf{r}_{k+1}$ is located on the LHS of $\mathbf{r}_k$, e.g.,

$$\begin{aligned}&x_k\left(y_k-v_{ty,k}+V_{o,k}T_s\sin\gamma_{o,k}\right)\\&\quad-y_k\left(x_k-v_{tx,k}+V_{o,k}T_s\cos\gamma_{o,k}\right)>0.\end{aligned}\tag{41}$$

Inside sector ABC that excludes lines $\overline{AB}$, $\overline{AC}$, it is clear that

$$\mathbf{v}_{o,k}^T\cdot\mathbf{r}_{k+1}<0.\tag{42}$$

Note that

$$\begin{aligned}\mathbf{v}_{o,k}^T\cdot\mathbf{r}_{k+1}/(V_{o,k}T_s)&=\cos\gamma_{o,k}\left(x_k-v_{tx,k}+V_{o,k}T_s\cos\gamma_{o,k}\right)\\&\quad+\sin\gamma_{o,k}\left(y_k-v_{ty,k}+V_{o,k}T_s\sin\gamma_{o,k}\right)\end{aligned}\tag{43}$$

Substituting (40)–(43) into (39) yields $\partial J/\partial V_{o,k}>0$. This implies that when $\mathbf{v}_{o,k}\in\Omega_2$, the cost function J monotonically increases with the increase of observer's velocity $V_{o,k}$.

Condition 2: When $\mathbf{v}_{o,k}\in\Omega_3$, one can imply that

$$y_k\cos\gamma_{o,k}-x_k\sin\gamma_{o,k}\geq 0\tag{44}$$

where the equality holds when $\mathbf{v}_{o,k}=\overrightarrow{AF}$, and $\mathbf{r}_{k+1}$ is located on the RHS of $\mathbf{r}_k$, e.g.,

$$\begin{aligned}&x_k\left(y_k-v_{ty,k}+V_{o,k}T_s\sin\gamma_{o,k}\right)\\&\quad-y_k\left(x_k-v_{tx,k}+V_{o,k}T_s\cos\gamma_{o,k}\right)<0.\end{aligned}\tag{45}$$

Inside sector ADE that excludes line $\overline{AD}$, it is clear that

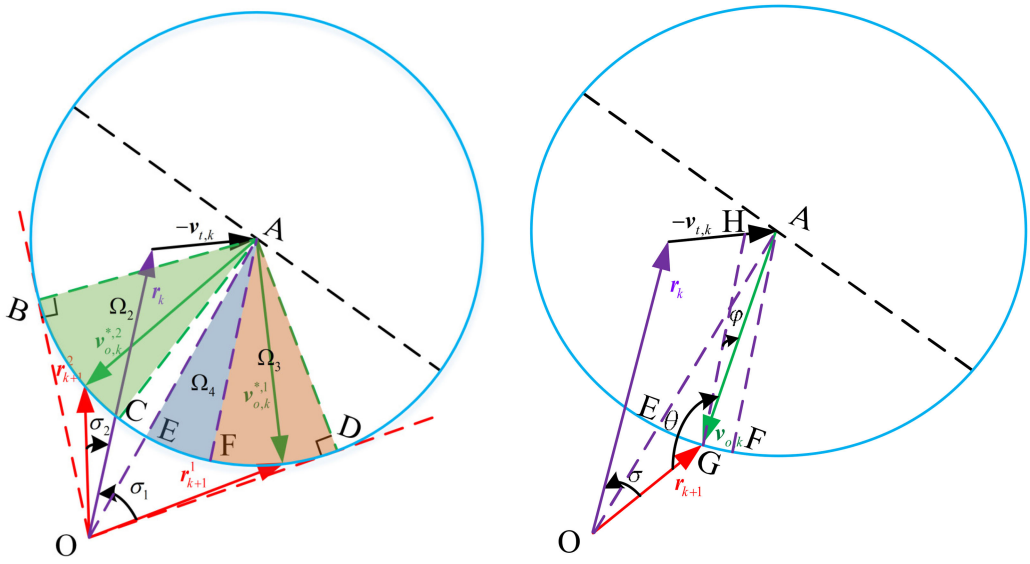
$$\mathbf{v}_{o,k}^T\cdot\mathbf{r}_{k+1}<0.\tag{46}$$

Substituting (43)–(46) into (39) yields $\partial J/\partial V_{o,k}>0$. This implies that when $\mathbf{v}_{o,k}\in\Omega_3$, the cost function J monotonically increases with the increase of observer's velocity $V_{o,k}$.

Condition 3: When $\mathbf{v}_{o,k}\in\Omega_4$, the geometric illustration is presented in Fig. 5(b), where line $\overline{GH}$ is parallel to $\mathbf{r}_k$. In this condition, it is clear that $\mathbf{v}_{o,k}$ is located on the LHS of $\mathbf{r}_k$ and

$$\begin{aligned}\frac{\partial J}{\partial V_{o,k}}=&-2T_s\frac{x_k\left(y_k-v_{ty,k}+V_{o,k}T_s\sin\gamma_{o,k}\right)-y_k\left(x_k-v_{tx,k}+V_{o,k}T_s\cos\gamma_{o,k}\right)}{\left[\left(x_k-v_{tx,k}+V_{o,k}T_s\cos\gamma_{o,k}\right)^2+\left(y_k-v_{ty,k}+V_{o,k}T_s\sin\gamma_{o,k}\right)^2\right]^3}\\&\times\left\{\left[\left(x_k-v_{tx,k}+V_{o,k}T_s\cos\gamma_{o,k}\right)^2+\left(y_k-v_{ty,k}+V_{o,k}T_s\sin\gamma_{o,k}\right)^2\right]\left(y_k\cos\gamma_{o,k}-x_k\sin\gamma_{o,k}\right)\right.\\&+2\left[x_k\left(y_k-v_{ty,k}+V_{o,k}T_s\sin\gamma_{o,k}\right)-y_k\left(x_k-v_{tx,k}+V_{o,k}T_s\cos\gamma_{o,k}\right)\right]\\&\left.\times\left[\cos\gamma_{o,k}\left(x_k-v_{tx,k}+V_{o,k}T_s\cos\gamma_{o,k}\right)+\sin\gamma_{o,k}\left(y_k-v_{ty,k}+V_{o,k}T_s\sin\gamma_{o,k}\right)\right]\right\}\end{aligned}\tag{39}$$

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(a) (b)

Fig. 5. Geometric illustration for the proof of Lemma 3 in the relative frame. (a) Three cases are considered to prove J is a monotonically increasing function with respect to $V_{o,k}$ when $\mathbf{v}_{o,k} \in \Omega_2, \Omega_3, \Omega_4$ with Ω_3 being sector ADF that excludes line $\overline{AD}$ and Ω_4 being sector AEF that excludes line $\overline{AF}$. In this figure, line $\overline{AF}$ is parallel to $\mathbf{r}_k$. (b) Proof of the geometric relationship shown in (50) when $\mathbf{v}_{o,k} \in \Omega_4$. In this figure, both lines $\overline{AF}$ and $\overline{GH}$ are parallel to $\mathbf{r}_k$.

$\mathbf{r}_{k+1}$ is located on the RHS of $\mathbf{r}_k$. Therefore, we have

$$-\|\mathbf{r}_k\|\,\|\mathbf{v}_{o,k}\|\sin\varphi = V_{o,k}T_s\left(y_k\cos\gamma_{o,k} - x_k\sin\gamma_{o,k}\right) \tag{47}$$

$$-\|\mathbf{r}_k\|\,\|\mathbf{r}_{k+1}\|\sin\sigma = x_k\left(y_k - v_{ty,k} + V_{o,k}T_s\sin\gamma_{o,k}\right) - y_k\left(x_k - v_{tx,k} + V_{o,k}T_s\cos\gamma_{o,k}\right). \tag{48}$$

Substituting (47) and (48) into (39) and after some algebraic manipulations, we get

$$\frac{\partial J}{\partial V_{o,k}} = -2\frac{\|\mathbf{r}_k\|^2\,\|\mathbf{v}_{o,k}\|\sin\sigma}{V_{o,k}\|\mathbf{r}_{k+1}\|^3}\left(\sin\varphi + 2\sin\sigma\cos\theta\right) \tag{49}$$

From Fig. 5(b), one can easily verify that

$$2\sin\sigma\cos\theta = -\sin\left(2\sigma-\varphi\right) - \sin\varphi. \tag{50}$$

Therefore, we have

$$\sin\varphi + 2\sin\sigma\cos\theta = -\sin\left(2\sigma-\varphi\right). \tag{51}$$

From Fig. 5(b), one can note that φ monotonically increases and σ monotonically decreases when $\mathbf{v}_{o,k}$ rotates from $\overrightarrow{AF}$ to $\overrightarrow{AE}$. Define $\bar{\sigma} = \sigma(\overrightarrow{AF})$ and $\hat{\sigma} = \sigma(\overrightarrow{AE})$. Then, we have $\hat{\sigma} \le 2\sigma - \varphi \le 2\bar{\sigma}$. Note that $0 < \sigma < \pi/2$, as discussed earlier, one can imply that $0 < 2\sigma - \varphi < \pi$, which means $\sin(2\sigma-\varphi) > 0$. With this in mind and substituting (51) into (49) gives $\partial J/\partial V_{o,k} > 0$. This implies that when $\mathbf{v}_{o,k} \in \Omega_4$, the cost function J monotonically increases with the increase of observer's velocity $V_{o,k}$. Finally, combining Conditions 1–3 leads to the proof of Lemma 3. ■

By using the results of Lemmas 1–3, the solution of CTO_2 is obtained in Theorem 1.

Theorem 1: The optimal observer maneuver, without any heading constraints, that maximizes cost function J is given by $\mathbf{v}_{o,k} = [V_{o,\max}T_s\cos\gamma^*_{o,k}, V_{o,\max}T_s\sin\gamma^*_{o,k}]^T$ where $\gamma^*_{o,k} = \max_{\gamma_{o,k}\in\{\gamma^{*,1}_{o,k},\gamma^{*,2}_{o,k}\}} J(\gamma_{o,k})$ with

$$\gamma^{*,1}_{o,k} = \arcsin\left(-\frac{2V_{o,\max}T_s\mathbf{r}_k^T\cdot\bar{\mathbf{r}}_k}{\sqrt{a^2+b^2}}\right) - \vartheta \tag{52}$$

$$\gamma^{*,2}_{o,k} = \pi - \arcsin\left(-\frac{2V_{o,\max}T_s\mathbf{r}_k^T\cdot\bar{\mathbf{r}}_k}{\sqrt{a^2+b^2}}\right) - \vartheta. \tag{53}$$

Here, $\sin\vartheta = b/\sqrt{a^2+b^2}$ and $\cos\vartheta = a/\sqrt{a^2+b^2}$, where

$$a = \|\mathbf{r}_k\|^2(y_k - 2v_{ty,k}) + 2x_kv_{tx,k}v_{ty,k} + y_k\left[v^2_{ty,k} - v^2_{tx,k} + (V_{o,\max}T_s)^2\right] \tag{54}$$

$$b = \|\mathbf{r}_k\|^2(x_k - 2v_{tx,k}) + 2y_kv_{tx,k}v_{ty,k} + x_k\left[v^2_{tx,k} - v^2_{ty,k} + (V_{o,\max}T_s)^2\right]. \tag{55}$$

Proof: From Lemmas 1–3, the proof of Theorem 1 is straightforward. ■

Remark 9: Theorem 1 mentions that cost function J is maximized when the robot moves with its maximum velocity $V_{o,\max}$. However, it is often not desirable to set the robot velocity as its maximum permissible value due to physical and operational reasons such as endurance consideration. In practice, the robot velocity is usually predefined in consideration of those constraints and mission objectives. Therefore, we limit the following analysis to the case where the robot velocity is constant as $V_{o,k} = V_o$.

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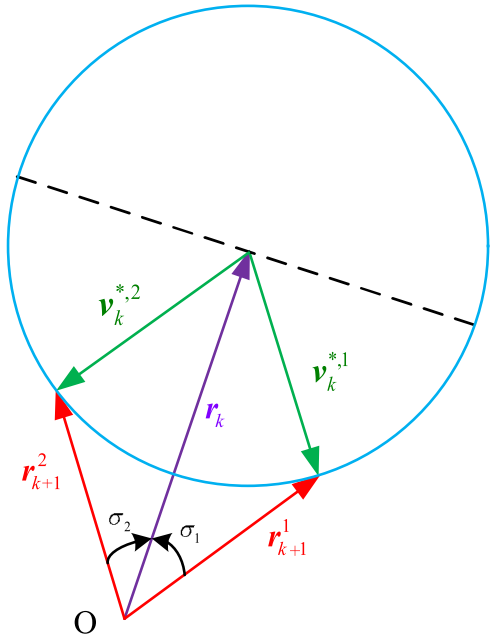


Fig. 6. Geometric interpretation of optimal maneuvers for stationary target in the relative frame with $\sigma_1 = \sigma_2$ and $\left\|\mathbf{r}_{k+1}^1\right\| = \left\|\mathbf{r}_{k+1}^2\right\|$.

B. Special Case: Stationary Target Localization

For stationary target, we have $\mathbf{r}_k = \bar{\mathbf{r}}_k$ and

$$a = \left[\|\mathbf{r}_k\|^2 + (V_o T_s)^2\right] y_k \tag{56}$$

$$b = \left[\|\mathbf{r}_k\|^2 + (V_o T_s)^2\right] x_k. \tag{57}$$

Then, condition given in (29) becomes

$$\sin(\gamma_{o,k} + \beta) = -\frac{2V_o T_s \|\mathbf{r}_k\|}{\|\mathbf{r}_k\|^2 + (V_o T_s)^2} \tag{58}$$

where $\sin\beta = x_k / \|\mathbf{r}_k\|$, $\cos\beta = y_k / \|\mathbf{r}_k\|$. Therefore, the optimal heading angle at time step k for stationary target localization is given by

$$\gamma_{o,k}^{*,1} = \arcsin\left(-\frac{2V_o T_s \|\mathbf{r}_k\|}{\|\mathbf{r}_k\|^2 + (V_o T_s)^2}\right) - \beta \tag{59}$$

$$\gamma_{o,k}^{*,2} = \pi - \arcsin\left(-\frac{2V_o T_s \|\mathbf{r}_k\|}{\|\mathbf{r}_k\|^2 + (V_o T_s)^2}\right) - \beta. \tag{60}$$

The geometric interpretation of optimal maneuvers is shown in Fig. 6. Unlike the moving target case, the two candidate solutions $\gamma_{o,k}^{*,1}$ and $\gamma_{o,k}^{*,2}$ are symmetric with respect to $\mathbf{r}_k$.

For moving target, it is clear from Fig. 3(b) that the change of the relative distance depends on target movement. However, when localizing a stationary target, the relative distance is monotonically decreasing as time goes. Also, the rate of the relative range change is proportional to the square of the observer velocity V_o^2 and the sampling time T_s, but inversely proportional to the current relative range $\|\mathbf{r}_k\|$. This result is given by the following proposition.

Proposition 1: For stationary target localization, the rate of the relative range between the observer and the target generated by the obtained optimal maneuver is given by

$$\frac{d\|\mathbf{r}_k\|}{dt} = -\frac{3V_o^2 T_s}{2\|\mathbf{r}_k\|}. \tag{61}$$

Proof: By definition, the rate of the relative range is determined by

$$\begin{aligned}\frac{d\|\mathbf{r}_k\|}{dt} &= \frac{\|\mathbf{r}_{k+1}\| - \|\mathbf{r}_k\|}{T_s} \\ &= \frac{\sqrt{\|\mathbf{r}_k\|^2 + (V_o T_s)^2 + 2V_o T_s\left(x_k \cos\gamma_{o,k}^* + y_k \sin\gamma_{o,k}^*\right)} - \|\mathbf{r}_k\|}{T_s}.\end{aligned} \tag{62}$$

Substituting (58) into (62) yields

$$\begin{aligned}\frac{d\|\mathbf{r}_k\|}{dt} &= \frac{\sqrt{\|\mathbf{r}_k\|^2 + (V_o T_s)^2 - \frac{4(V_o T_s)^2\|\mathbf{r}_k\|^2}{\|\mathbf{r}_k\|^2 + (V_o T_s)^2}} - \|\mathbf{r}_k\|}{T_s} \\ &= \frac{\sqrt{\|\mathbf{r}_k\|^2 + \frac{(V_o T_s)^2\left[(V_o T_s)^2 - 3\|\mathbf{r}_k\|^2\right]}{\|\mathbf{r}_k\|^2 + (V_o T_s)^2}} - \|\mathbf{r}_k\|}{T_s}.\end{aligned} \tag{63}$$

Since T_s is small, we have

$$\begin{aligned}\frac{d\|\mathbf{r}_k\|}{dt} &\approx \frac{\sqrt{\|\mathbf{r}_k\|^2 - 3(V_o T_s)^2} - \|\mathbf{r}_k\|}{T_s} \\ &= \frac{\|\mathbf{r}_k\|\sqrt{1 - \frac{3(V_o T_s)^2}{\|\mathbf{r}_k\|^2}} - \|\mathbf{r}_k\|}{T_s} \\ &\approx \frac{\|\mathbf{r}_k\|\left(1 - \frac{3}{2}\frac{(V_o T_s)^2}{\|\mathbf{r}_k\|^2}\right) - \|\mathbf{r}_k\|}{T_s} \\ &= -\frac{3V_o^2 T_s}{2\|\mathbf{r}_k\|}\end{aligned} \tag{64}$$

which clearly indicates the that relative distance is monotonically decreasing as time goes. ■

C. Optimal Maneuver With Heading Constraint

Theorem 1 reveals that, given the observer velocity, there might exist two optimal heading angles, $\gamma_{o,k}^{*,1}$ and $\gamma_{o,k}^{*,2}$, which generate the same value of cost function for the observer to minimize target localization error. From the standpoint of real application, it is desirable to avoid zigzag heading angle change to maintain the physical constraints. Therefore, it is necessary to consider physical constraints and select one maneuver vector respecting those constraints, when implementing Theorem 1 in practice. This section will show how the heading angle constraint (2) can be embedded into the proposed optimal solution.

Let $\gamma_{o,k-1}^*$ be the optimal heading angle at the previous time step. Without loss of generality, assume that $\gamma_{o,k}^{*,1}$ is closer to $\gamma_{o,k-1}^*$ than $\gamma_{o,k}^{*,2}$, as shown in Fig. 7, e.g., $|\gamma_{o,k}^{*,1} - \gamma_{o,k-1}^*| \leq |\gamma_{o,k}^{*,2} - \gamma_{o,k-1}^*|$. Then, the solution of CTO_1 is given by the following theorem.

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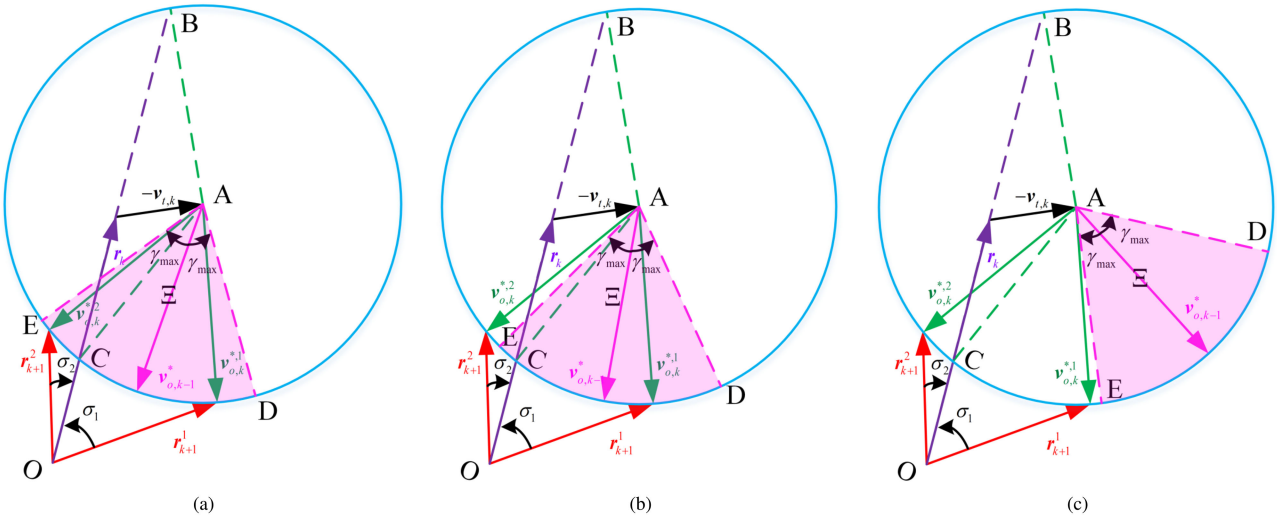


Fig. 7. Geometric illustration for moving target localization with turning rate limit in the relative frame. The magenta sector ADE, denoted as Ξ, quantifies the maximum permissible movement region of the robot at current time instant. (a) Case 1: both $\gamma_{o,k}^{*,1}$ and $\gamma_{o,k}^{*,2}$ are located inside sector ADE. (b) Case 2: only $\gamma_{o,k}^{*,1}$ is located inside sector ADE. (c) Case 3: $\gamma_{o,k}^{*,1}$ is located outside sector ADE.

Theorem 2: Let Ξ be sector ADE (shown in Fig. 7). Then, we have the following results.

1) If $\gamma_{o,k}^{*,1} \in \Xi$ and $\gamma_{o,k}^{*,2} \in \Xi$ [refer to Fig. 7(a)], then, the optimal observer heading angle $\gamma_{o,k}^{*}$ considering turning rate constraint is determined as

$$\gamma_{o,k}^{*} = \begin{cases} \gamma_{o,k}^{*,1}, & \text{if} \quad J\left(\gamma_{o,k}^{*,1}\right) = J\left(\gamma_{o,k}^{*,2}\right) \\ \max\limits_{\gamma_{o,k} \in \left\{\gamma_{o,k}^{*,1}, \gamma_{o,k}^{*,2}\right\}} J\left(\gamma_{o,k}\right), & \text{otherwise} \end{cases}. \tag{65}$$

2) If only $\gamma_{o,k}^{*,1} \in \Xi$ [refer to Fig. 7(b)], then, the optimal observer heading angle $\gamma_{o,k}^{*}$ considering turning rate constraint is determined as

$$\gamma_{o,k}^{*} = \max_{\gamma_{o,k} \in \left\{\gamma_{o,k}^{*,1}, \bar{\gamma}_{o,k}^{*,1}, \bar{\gamma}_{o,k}^{*,2}\right\}} J\left(\gamma_{o,k}\right) \tag{66}$$

where $\bar{\gamma}_{o,k}^{*,1} = \gamma_{o,k-1}^{*} - \gamma_{\max}$ and $\bar{\gamma}_{o,k}^{*,2} = \gamma_{o,k-1}^{*} + \gamma_{\max}$.

3) If $\gamma_{o,k}^{*,1} \notin \Xi$ [refer to Fig. 7(c)], then, the optimal observer heading angle $\gamma_{o,k}^{*}$ considering turning rate constraint is determined as

$$\gamma_{o,k}^{*} = \max_{\gamma_{o,k} \in \left\{\bar{\gamma}_{o,k}^{*,1}, \bar{\gamma}_{o,k}^{*,2}\right\}} J\left(\gamma_{o,k}\right). \tag{67}$$

Proof: When $\gamma_{o,k}^{*,1} \in \Xi$ and $\gamma_{o,k}^{*,2} \in \Xi$, the optimal heading angle that maximizes cost function J is obviously given by $\gamma_{o,k}^{*} = \max_{\gamma_{o,k} \in \left\{\gamma_{o,k}^{*,1}, \gamma_{o,k}^{*,2}\right\}} J\left(\gamma_{o,k}\right)$. If these two candidate heading solutions provide the same level of optimality, we then choose the solution which is closer to the previous heading angle to avoid large heading change, as shown in (65).

Lemma 1 indicates that if $\mathbf{v}_{o,k} = \overrightarrow{AB}$ or $\mathbf{v}_{o,k} = \overrightarrow{AC}$, the cost function J achieves the minimum value as $J(\overrightarrow{AB}) = 0$ and $J(\overrightarrow{AC}) = 0$. Also, when $\mathbf{v}_{o,k} = \mathbf{v}_{o,k}^{*,1}$ or $\mathbf{v}_{o,k} = \mathbf{v}_{o,k}^{*,2}$, the cost function J is locally maximized. Therefore, the cost function J monotonically decreases when the observer maneuver vector $\mathbf{v}_{o,k}$ rotates from $\mathbf{v}_{o,k}^{*,1}$ (or $\mathbf{v}_{o,k}^{*,2}$) to $\overrightarrow{AB}$ or to $\overrightarrow{AC}$ since there is no solution $\partial J / \partial \gamma_{o,k} = 0$ that is located between $\mathbf{v}_{o,k}^{*,1}$ (or $\mathbf{v}_{o,k}^{*,2}$) and $\overrightarrow{AB}$ or $\mathbf{v}_{o,k}^{*,1}$ (or $\mathbf{v}_{o,k}^{*,2}$) and $\overrightarrow{AC}$. This leads to the proof of results 2) and 3) in Theorem 2. ■

Remark 10: Note that when $\gamma_{o,k}^{*,2}$ is closer to $\gamma_{o,k-1}^{*}$ than $\gamma_{o,k}^{*,1}$, i.e., $|\gamma_{o,k}^{*,1} - \gamma_{o,k-1}^{*}| > |\gamma_{o,k}^{*,2} - \gamma_{o,k-1}^{*}|$, similar results as shown in Theorem 2 can also be obtained.

D. Algorithm Summary

By summarizing the results presented in Section IV-A and Section IV-C, the proposed optimal trajectory optimization algorithm for target localization with bearing-only measurement is summarized in Algorithm 1.

E. Observability Analysis Under the Proposed Algorithm

In the development of the proposed algorithm, the assumption that $\mathbf{r}_k$ is available is utilized in trajectory optimization. However, the information on $\mathbf{r}_k$ is extracted from Kalman filter in practice. This means that only when the system is observable, $\mathbf{r}_k$ is available to the observer. For this reason, this section will analyze the system observability under the proposed approach using geometric concepts. The primary objective of observability analysis is to find out whether or not $\mathbf{r}_k$ can be uniquely determined within a finite time period from time instant k onward. The results are given in the following proposition.

Proposition 2: For constant-maneuvering target, the target observability is ensured under the proposed algorithm.

Proof: In [13], Song revealed that the necessary and sufficient condition to guarantee system observability for constant-maneuvering target with bearing-only measurement is given by

$$\mathbf{r}(t) \neq \lambda(t) \begin{bmatrix} a_{11} + a_{12}\Delta t + a_{13}\frac{\Delta t^2}{2} \\ a_{21} + a_{22}\Delta t + a_{23}\frac{\Delta t^2}{2} \end{bmatrix} \tag{68}$$

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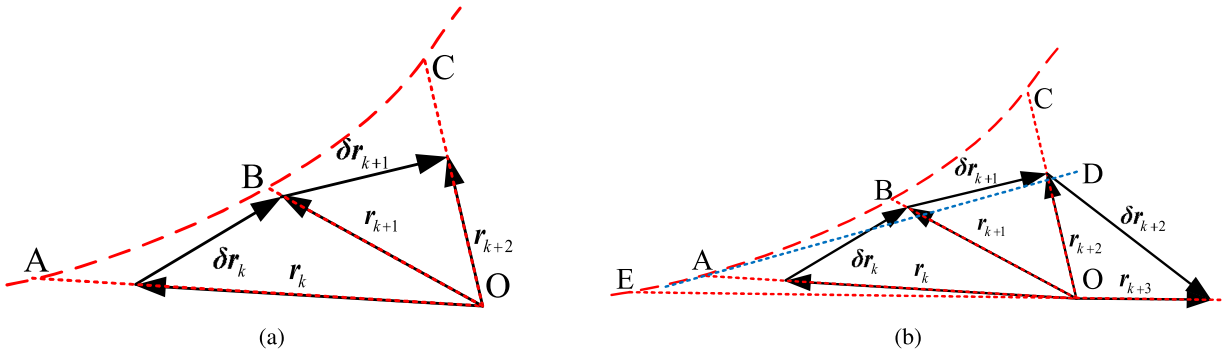


Fig. 8. (a) Geometric interpretation of the necessary and sufficient observability condition for three consecutive time steps for constant-maneuvering target with bearing-only measurement. (b) Geometric interpretation utilized in the proof of Proposition 2. In (b), line $\overline{AD}$ is the tangential line of quadratic curve AC at point A.

Algorithm 1: Optimal Trajectory Optimization for Target Localization With Bearing-Only Measurement.

Input: Estimated target position vector $\mathbf{r}_{t,k}$, estimated target manoeuvre vector $\mathbf{v}_{t,k}$, previous observer manoeuvre vector $\mathbf{v}_{o,k-1}$, maximum allowable heading angle change $\gamma_{\max}$
Output: Optimal observer heading angle $\gamma^*_{o,k}$

```
Calculate the candidate optimal heading angles γ*,1_{o,k} and γ*,2_{o,k} using Eqs. (17) and (18)
if |γ*,1_{o,k} − γ*_{o,k−1}| ≤ |γ*,2_{o,k} − γ*_{o,k−1}| then
    γ̄*_{o,k} = γ*,1_{o,k}
else
    γ̄*_{o,k} = γ*,2_{o,k}
end if
if |γ*,1_{o,k} − γ*_{o,k−1}| ≤ γ_max and |γ*,2_{o,k} − γ*_{o,k−1}| ≤ γ_max then
    γ*_{o,k} is given by Eq. (65)
else if |γ̄*_{o,k} − γ*_{o,k−1}| ≤ γ_max then
    γ*_{o,k} is given by Eq. (66)
else
    γ*_{o,k} is given by Eq. (67)
end if
```

for some $t \in (t_0, t_f]$ with t_0 being the initial time, where $\Delta t = t - t_0$ and $\lambda(t)$ is an arbitrary scalar function. The coefficients a_{ij} in (68) are arbitrary constants but not all zero.

According to condition (68), if the system is unobservable, there exists a quadratic curve such that the relative position vector has a linear relationship with the quadratic curve. A geometric illustration of condition (68) for three consecutive time steps is presented in Fig. 8. If the system is unobservable, there exist an arbitrary scalar function $\lambda(t)$ and a quadratic curve AC such that $\mathbf{r}_k = \lambda(t_k)\overrightarrow{OA}$, $\mathbf{r}_{k+1} = \lambda(t_{k+1})\overrightarrow{OB}$ and $\mathbf{r}_{k+2} = \lambda(t_{k+2})\overrightarrow{OC}$ locates on the quadratic curve AC. It is clear that $\sigma = 0$ is a special case that violates condition (68).

Define $r_x(t) = a_{11} + a_{12}\Delta t + a_{13}\frac{\Delta t^2}{2}$ and $r_y(t) = a_{21} + a_{22}\Delta t + a_{23}\frac{\Delta t^2}{2}$. Due to the property of quadratic polynomials, there exists a certain time instant $t_* > t_0$ such that both $r_x(t)$ and $r_y(t)$ are monotonic functions when $t > t_*$. Without loss of generality, we assume that $t_k > t_*$. From previous analysis, we know that the proposed algorithm generates a unique solution that forces the relative position vector rotate clockwise or anticlockwise around the target. Without loss of generality, the clockwise rotation direction [as shown in Fig. 8(a)] is utilized in the following analysis. With this in mind, one can imply that there exists a certain time instant (assuming this occurs at time step $k+3$ for simplicity) such that $\mathbf{r}_{k+3}$ is located on the right hand of $\overrightarrow{AD}$, as shown in Fig. 8(b), where line $\overline{OD}$ is the tangential line of quadratic curve AC at point A. Since quadratic curves are convex, it is easy to verify that the only intersection point between quadratic curve AC and $\mathbf{r}_{k+3}$ is E. This apparently violates the condition that $r_x(t)$ and $r_y(t)$ are monotonic when $t > t_*$. Therefore, $\forall t_k$, there exists some $t \in (t_k, t_f]$ such that condition (68) is satisfied. This means that target is observable under the proposed approach. ■

V. Simulation Studies

In this section, estimator-in-the-loop simulations are performed to validate the proposed optimal trajectory optimization algorithm. To illustrate and evaluate performance characteristics, the proposed algorithm was tested and compared with the algorithms proposed in [14] and [42], via extensive simulations. This section presents representative results for three typical tracking scenarios.

A. Simulation Setup

In the given scenarios, an aerial robot tracks a ground target using bearing-only measurement. The robot is assumed to be flying with a constant velocity (CV) $V_o = 20\,\text{m/s}$. The initial position of the robot is $(20\,\text{m}, 50\,\text{m})$ and the target initially locates at $(0\,\text{m}, 0\,\text{m})$. The turning rate of the robot is constrained by $\omega_{\max} = 1.5\,\text{rad/s}$ and the sampling time is set as $T_s = 0.1\,\text{s}$. The angle measurement noise ε is assumed to be zero-mean white Gaussian as $\mathcal{N}(\varepsilon; 0, \sigma_\varepsilon^2)$ with $\sigma_\varepsilon = 0.5°$. In the simulation scenarios, three different types of targets are considered.

Case 1: Target is stationary.

Case 2: Target moves in a straight line with CV $V_t = 5.83\,\text{m/s}$ and constant heading $\gamma_t = 0.54\,\text{rad}$.

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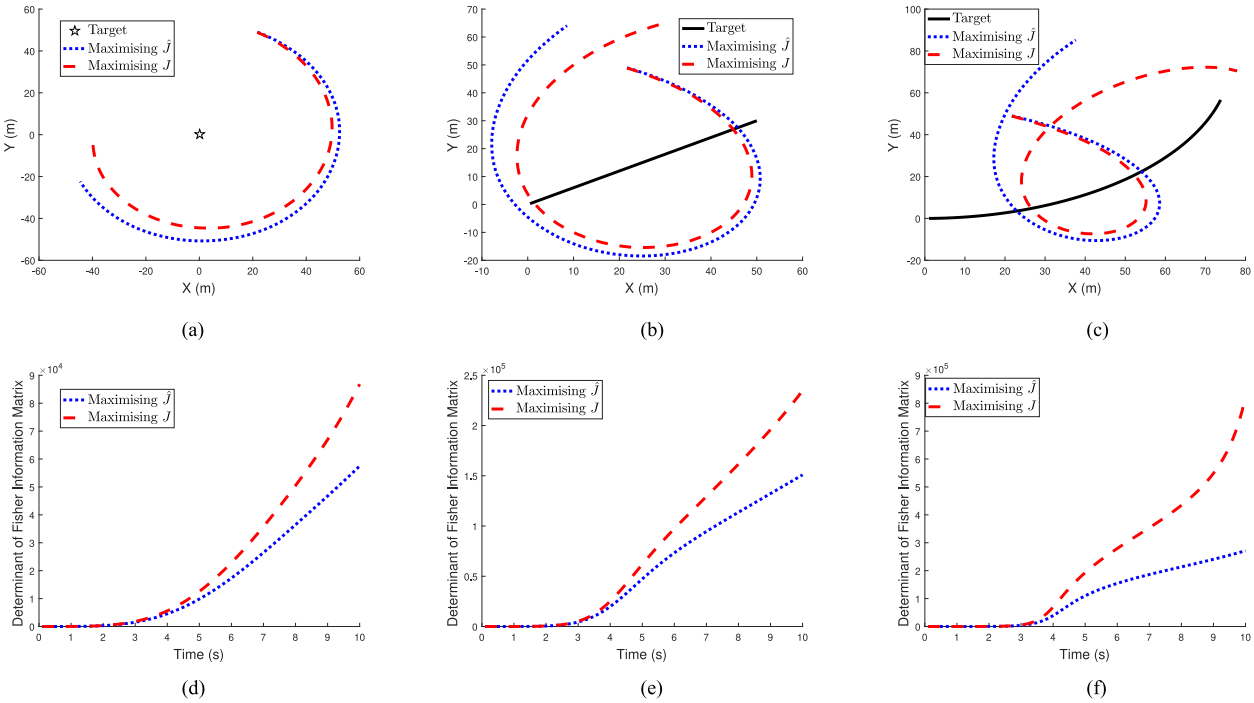


Fig. 9. Simulation results obtained from the proposed algorithm and [14] for three different target motions. The first column corresponds to stationary target, the second column refers to nonmaneuvering target, and the third column is for constant-maneuvering target. (a) Robot and target trajectories of Case 1. (b) Robot and target trajectories of Case 2. (c) Robot and target trajectories of Case 3. (d) Determinant of FIM of Case 1. (e) Determinant of FIM of Case 2. (f) Determinant of FIM of Case 3.

Case 3: Target moves with CV $V_t = 10\,\text{m/s}$, constant turning rate $\dot{\gamma}_t = \pi/24\,\text{rad/s}$ and initial heading $\gamma_{t,1} = 0^\circ$.

To implement the proposed trajectory optimization algorithm, the required information on target position and velocity are estimated using the well-known extended Kalman filter (EKF). In EKF design, as we utilize the piece-wise nonmaneuvering assumption in algorithm development, the standard CV model is utilized as the target dynamics for prediction. Let $\mathbf{x}_k = [x_{t,k}, \dot{x}_{t,k}, y_{t,k}, \dot{y}_{t,k}]^T$, the state transition of CV model is determined by

$$\mathbf{x}_k = F_{\text{CV}}\mathbf{x}_{k-1} + G_{\text{CV}}w_{k-1} \tag{69}$$

with

$$F_{\text{CV}} \triangleq \mathbb{I}_{2\times 2} \otimes \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}, \quad G_{\text{CV}} \triangleq \begin{bmatrix} T_s^2/2 \\ T_s \\ T_s^2/2 \\ T_s \end{bmatrix} \tag{70}$$

where $\mathbb{I}_{2\times 2}$ denotes the 2×2 identity matrix, and $w_k \sim \mathcal{N}\left(\cdot; 0, \sigma_v^2\right)$ the Gaussian process noise with $\sigma_v = 0.5\,\text{m/s}^2$.

Since the estimation error covariance matrix is positive definite, its quadratic form constitutes a hyper uncertainty ellipsoid that determines the error distribution and the axes of ellipsoids are defined by the eigenvalues of the error covariance matrix. Therefore, the volume of the uncertainty ellipsoid can be used as a meaningful performance metric to quantify the estimation accuracy. Let P denote the target position error covariance, the one sigma area of the uncertainty ellipsoid is determined as

$$A_{1\sigma} = \pi\sqrt{|P|}. \tag{71}$$

Note that the FIM is inversely related with the error covariance matrix. Thus, the minimization of the volume of the uncertainty ellipsoid can be achieved by maximizing the determinant of FIM. This means that the robot trajectory that generates higher observability of the target has higher value of $|F|$. Therefore, we utilize the determinant of FIM as a metric to check the performance in simulations. Note that the FIM can be calculated in a recursive way as described in [44].

B. Comparison With [14]

By utilizing a lower bound of the determinant of the incremental of FIM as a cost function, Anjaly and Ratnoo [14] suggested a solution to minimize estimation error for bearing-only measurement navigation. This section compares the proposed approach with [42] to demonstrate the advantages of the proposed geometric measure. The one-step optimal observer maneuver that maximizes $\hat{J}$ is obtained through the genetic algorithm (GA) solver from MATLAB Optimization Toolbox since it is difficult to find analytical solutions that maximize $\hat{J}$. For fair comparison, the robot heading constraint is also embedded into the GA solver. It should be pointed out [14] never considered the physical constraints in the implementation. Note that for one-step trajectory optimization, maximizing $(\sigma/\|\mathbf{r}_k\|)^2$ is equivalent to maximize the absolute value of the separation angle between two consecutive time steps since $\|\mathbf{r}_k\|$ is known.

The comparison results obtained by these two different algorithms for all three cases are shown in Fig. 9. The first row in Fig. 9 provides the robot and target trajectories for these three cases. From Fig. 9(a)–(c), it can be noted that the robot

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TABLE I
RMSE COMPARISONS OVER1000 MONTE-CARLO RUNS

	Metric	Case 1	Case 2	Case 3
Optimising $\hat{J}$ [14]	Mean	$1.6259\,m$	$1.8264\,m$	$2.5933\,m$
	Std	$0.4171\,m$	$0.5852\,m$	$0.6319\,m$
Optimising J (11)	Mean	$1.5027\,m$	$1.5408\,m$	$2.0562\,m$
	Std	$0.4059\,m$	$0.5466\,m$	$0.6412\,m$

TABLE II
RMSE COMPARISONS OVER1000 MONTE-CARLO RUNS

	Metric	Case 1	Case 2	Case 3
Compared approach [42]	Mean	$1.5031\,m$	$1.5472\,m$	$2.1336\,m$
	Std	$0.7319\,m$	$0.8011\,m$	$1.0338\,m$
Proposed approach	Mean	$1.5027\,m$	$1.5408\,m$	$2.0562\,m$
	Std	$0.4059\,m$	$0.5466\,m$	$0.6412\,m$

trajectories generated by the proposed algorithm always locate inside the robot trajectories obtained by maximizing $\hat{J}$. This reveals that the proposed algorithm provides a proper balance between maximizing the absolute value of the separation angle and the minimization of the relative range, whereas maximizing $\hat{J}$ only maximizes the absolute value of the separation angle. The corresponding determinant of FIM obtained from two different algorithms are depicted in the second row of Fig. 9. It is clear that the proposed algorithm generates higher value of $|F_k|$ as time goes, demonstrating that optimization of the proposed geometric metric (11) generates higher system observability than that of maximizing $\hat{J}$. Since the one sigma volume of the uncertainty ellipsoid $A_{1\sigma}$ is inversely proportional to $|F|$, the proposed trajectory optimization algorithm is expected to generate smaller error ellipsoid.

For better illustration, Monte-Carlo simulations are carried out to validate the proposed approach. In each Monte-Carlo run, the robot's initial position $(x_{o,1}, y_{o,1})$ is randomly sampled from uniform distributions as $x_{o,1} \sim \mathcal{U}(-40\,\mathrm{m}, 40\,\mathrm{m})$ and $y_{o,1} \sim \mathcal{U}(-40\,\mathrm{m}, 40\,\mathrm{m})$. The standard deviation (Std) of the measurement noise σ_ε at each Monte-Carlo run is also randomly sampled from uniform distribution as $\sigma_\varepsilon \sim \mathcal{U}(0.5^\circ, 2^\circ)$. The time averaged and Std of root-mean-square errors (RMSEs) over 1000 Monte-Carlo runs obtained from different algorithms for the aforementioned three cases are summarized in Table I. From this table, it is clear that the proposed analytical algorithm that maximizing J yields improved localization performance than numerically maximizing $\hat{J}$, especially when target is moving. This coincides with the results shown in the second row of Fig. 9. The Monte-Carlo simulation results also reveal that minimizing the relative range between the robot and the target provides the possibility to reduce the estimation uncertainty, which complies with the geometric concept discussed earlier. To further demonstrate that maximizing $\hat{J}$ is indeed different from J, t-test between these two approaches is performed with 5% significance level and the result is $h = 1$ for all three cases. This reveals that optimizing these two different performance metrics is statistically different. Also note from Table I that both algorithms generate comparable level of Std over large number of Monte-Carlo simulations. This can be attributed to the fact that we consider the physical turning rate limit in implementing both algorithms, which means the generated optimal heading command is unique and consistent.

C. Comparison With [42]

By minimizing the trace of the error covariance at next time step, an analytical optimal trajectory optimization algorithm for target localization with bearing-only measurement is proposed in [42], where the control input is robot's position. However, since physical constraints of the robot are not rigorously considered in this paper, the solution might become infeasible. This section compares the proposed approach with [42] to demonstrate the importance of considering turning rate constraint.

The simulation results obtained by these two different algorithms for two sample runs of Case 3 are depicted in Fig. 10, in which each row corresponds to one sample run. It is clear from the first column of Fig. 10 that analytical solution proposed in [42] finds multiple candidate optimal robot trajectories. Numerically picking one of them inevitably generates zigzag form motion trajectory, thus resulting in large heading angle change. This can be noted from the blue peaks of robot turning rate in the third column of Fig. 10. As a comparison, by considering the physical turning rate constraint, the proposed algorithm only generates one unique feasible optimal solution. When the candidate optimal heading angles generated by Theorem 1 locate outside the maximum permissible movement region, the final optimal solution is obtained by using the maximum turning rate, as shown in Theorem 2. This fact can be clearly observed from the zoomed-in figure in the third column of Fig. 10. Also, it follows from the second column of Fig. 10 that the determinant of FIM depends on robot's trajectory and one-step minimization of the trace of the error covariance does not always ensure higher value of $|F|$ than maximizing the proposed geometric cost function J.

To further evaluate the performance of these two different algorithms under various conditions, Monte-Carlo simulations with same random initializations, shown in Section V-B, are conducted. The time average and Std of RMSE for Cases 1–3 over 1000 Monte-Carlo runs are summarized in Table II. Similar to Section V-B, t-test is performed with 5% significance level to check the statistical difference and the result is $h = 0$ for all three cases. It was shown in [1] that the trace of error covariance matrix or the inverse of FIM, known as A-optimality criterion, captures the geometric system dependencies. Therefore, it is expected that maximizing the proposed cost function and minimizing the trace of error covariance will generate comparable performance under same conditions. This can be clearly observed from Table II and confirmed by the t-test results. However, it is clear from Table II that the Std of RMSE generated by algorithm [42] is larger than the proposed algorithm. The reason is that algorithm [42] might provide multiple solutions and will numerically pick up one solution from the solution set, which means that algorithm [42] might generate totally different trajectories with the same initial conditions, as we can note from the first column of Fig. 10. As a comparison, the proposed algorithm generates a unique and consistent solution

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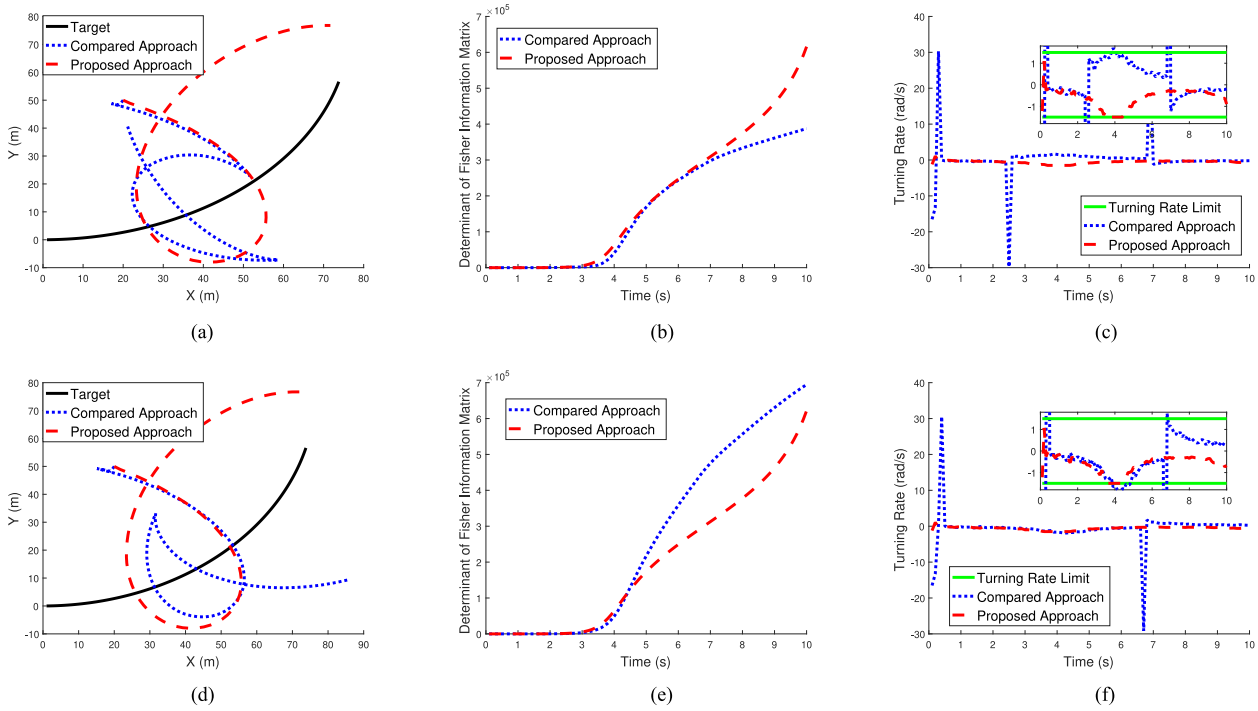


Fig. 10. Simulation results obtained from the proposed algorithm and [42] for constant-maneuvering target with each row corresponding to one sample run. The difference between these two rows is resulted from the multiple solution problem of [42]. (a) Robot and target trajectories. (b) Determinant of FIM. (c) Robot turning rate. (d) Robot and target trajectories. (e) Determinant of FIM. (f) Robot turning rate.

with the consideration of physical constraints. Therefore, the advantage of the proposed approach, compared to [42] is clear: the algorithm developed considers the turning rate limit in trajectory optimization. Therefore, the proposed algorithm is more practical and promising for real applications.

VI. Conclusion

The problem of analytical trajectory optimization for target localization with bearing-only measurement was studied in this paper. System observability was analyzed in the relative domain by using geometric conditions. This provided better understanding of how observability influences the estimation performance. By leveraging the geometric performance measure derived as the cost function, the optimal observer heading command was derived analytically and the maximum permissible turning rate was also considered in the proposed approach. Extensive numerical simulations with comparisons were performed to validate the analytical finds.

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